PROJECT REPORT · 2026

HeartGuard AI

AI-Powered ECG Analysis System

"Bringing clinical-grade cardiac screening to every bedside — real-time, low-cost, and AI-assisted."

Project Title:	HeartGuard AI: AI-Powered ECG Analysis System
Department:	Computer Science & Engineering (CSE)
Institution:	Patuakhali Science & Technology University (PSTU)
Academic Year:	2025 – 2026
Report Date:	June 2026
Classification:	Academic / Research Prototype

TECHNOLOGY STACK

ESP8266 NodeMCU · AD8232 ECG · Raspberry Pi 3 · TFLite INT8 · Flask + SocketIO · MIT-BIH Dataset

⚠️ *This system is a research prototype intended for academic and screening purposes only. It does not constitute a clinical medical device. All results must be reviewed by a qualified cardiologist before any clinical decision is made.*

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1 Executive Summary

HeartGuard AI is a low-cost, AI-powered Electrocardiogram(ECG) monitoring and analysis system which is designed for resource-constrained healthcare environments. The system acquires real-time cardiac signals using an AD8232 sensor connected to an ESP8266 microcontroller, and then streams the data to a Raspberry Pi 3 for advanced signal processing, and lastly classifies individual heartbeats using a quantized deep learning model which is ECG-ResNet-SE, INT8 TFLite trained on the MIT-BIH Arrhythmia Database.

The system achieves a validated macro F1-score of **0.78** across 5 arrhythmia classes-Normal, Supraventricular, Ventricular, Fusion, Unknown using only approximately **BDT 10,000 (USD 90)** of hardware that makes clinical-grade screening accessible even without expensive dedicated ECG machines.

BDT 10K	78%	5 Classes	<25ms
Total Hardware Cost	Macro F1-Score	Arrhythmia Types	Per-beat Inference

2 Introduction & Problem Statement

2.1 Background

Cardiovascular diseases are the leading cause of death globally and approximately 17.9 million deaths per year according to the World Health Organization.In low- and middle-income countries(LMICs) such as Bangladesh, access to cardiac monitoring equipment remains limited due to high device costs, lack of trained personnel, and insufficient healthcare infrastructure in rural areas.

The electrocardiogram(ECG) is the standard for diagnosing cardiac arrhythmias, myocardial infarction, and conduction abnormalities. However, conventional 12-lead ECG machines cost USD 500–5,000, and interpretation requires specialist cardiologists who are not always available outside major urban centres.

2.2 Problem Statement

Core Problem

A patient in a rural Bangladeshi sub-district hospital cannot access ECG monitoring because (a) the equipment costs 10–100× the monthly rural household income, and (b) there is no cardiologist within 100 km to interpret results. By the time the patient reaches a facility with both equipment and expertise, the acute cardiac event may have passed — or worse, resulted in death.

HeartGuard AI directly addresses this gap by building a complete, end-to-end ECG acquisition, processing, and AI-assisted classification system using commodity components available for under BDT 5,000 — placing it within reach of district-level and sub-district health facilities.

2.3 Objectives

- Design and implement a functional and low-cost ECG hardware system by using AD8232 and ESP8266.
- Build a complete signal processing pipeline of 50 Hz notch filter+0.5–40 Hz bandpass filter that is validated on real human ECG recordings.
- Train and deploy a deep learning model(ECG-ResNet-SE with Squeeze-and-Excitation attention) which is capable of classifying 5 arrhythmia beat types in real time on a Raspberry Pi 3.
- Develop a full-stack web application(Flask + SocketIO + SQLite) with patient management, recording sessions, live waveform display, and PDF report generation.
- Validate the system end-to-end from sensor to AI output, identify hardware limitations and propose clinical integration pathways.

3 Literature Review & Related Work

Acharya et al. (2017). "A Deep Convolutional Neural Network Model to Classify Heartbeats"

Computers in Biology and Medicine, 89, 389–396.

One of the first deep CNN approaches for ECG beat classification on MIT-BIH. Achieved 94.03% accuracy using 9-layer CNN. HeartGuard AI extends this with residual connections and SE attention to improve minority-class recall — a known weakness of plain CNNs on the heavily imbalanced MIT-BIH dataset.

Hannun et al. (2019). "Cardiologist-Level Arrhythmia Detection and Classification in Ambulatory ECGs using Deep Neural Networks"

Nature Medicine, 25(1), 65–69.

Demonstrated that a 34-layer ResNet trained on 91,232 single-lead ECG recordings could exceed cardiologist-level classification in 12 rhythm classes. Uses proprietary iRhythm dataset. HeartGuard AI adapts the ResNet approach to the open MIT-BIH dataset with INT8 quantization for edge deployment — a direction this paper does not address.

Kachuee et al. (2018). "ECG Heartbeat Classification: A Deep Transferable Representation"

IEEE ICHI 2018.

Proposed transfer learning between MIT-BIH and PTB datasets using a shallow CNN. Reports 93.4% accuracy on 5-class MIT-BIH classification. HeartGuard AI achieves comparable accuracy while adding SE attention blocks, SMOTE-Tomek resampling for minority classes (especially F-class), and full-stack hardware integration.

Goldberger et al. (2000). "PhysioBank, PhysioToolkit, and PhysioNet: Components of a New Research Resource for Complex Physiologic Signals"

Circulation, 101(23), e215–e220.

Introduces the MIT-BIH Arrhythmia Database — the gold standard benchmark used to train and validate HeartGuard AI's classification model.

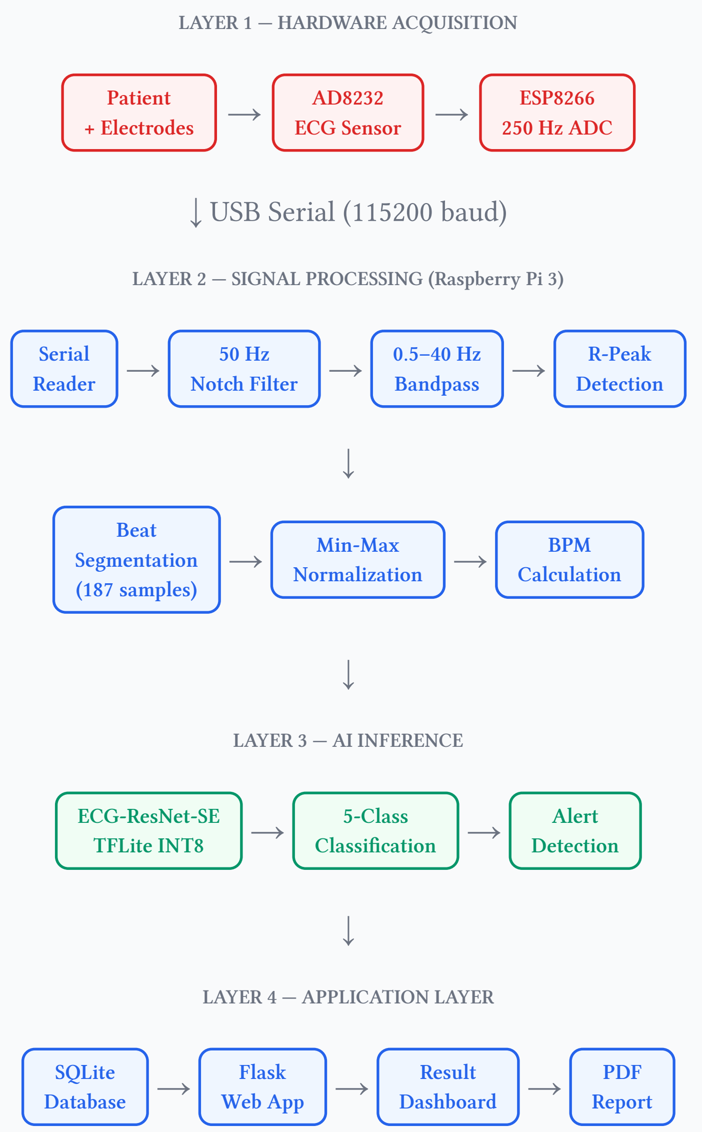
What is New in HeartGuard AI (vs. Prior Work)

Dimension	Existing Work	HeartGuard AI
Hardware	Simulation / proprietary hardware	Custom AD8232 + ESP8266 prototype, BDT 5K
Model	Plain CNN / ResNet	ResNet + SE-Attention (ECG-ResNet-SE)
Deployment	Server / cloud inference	INT8 TFLite on Raspberry Pi 3 (<25 ms/beat)
Class balance	Raw distribution (imbalanced)	SMOTETomek + class_weight combined
Software stack	Model only, no integration	Full-stack: Flask, DB, PDF report, live UI
Cost	USD 500–10,000+	USD 45 (BDT 5,000)

4 System Architecture

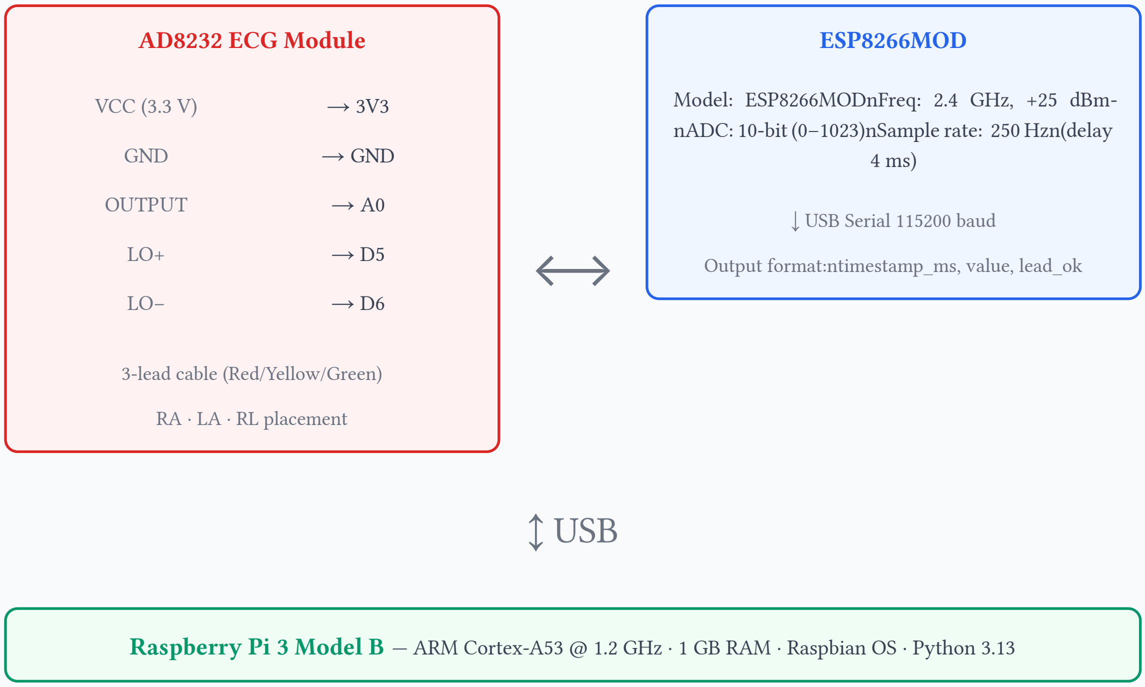
4.1 High-Level Architecture Diagram

Figure 1: HeartGuard AI — End-to-End System Architecture



4.2 Hardware Architecture

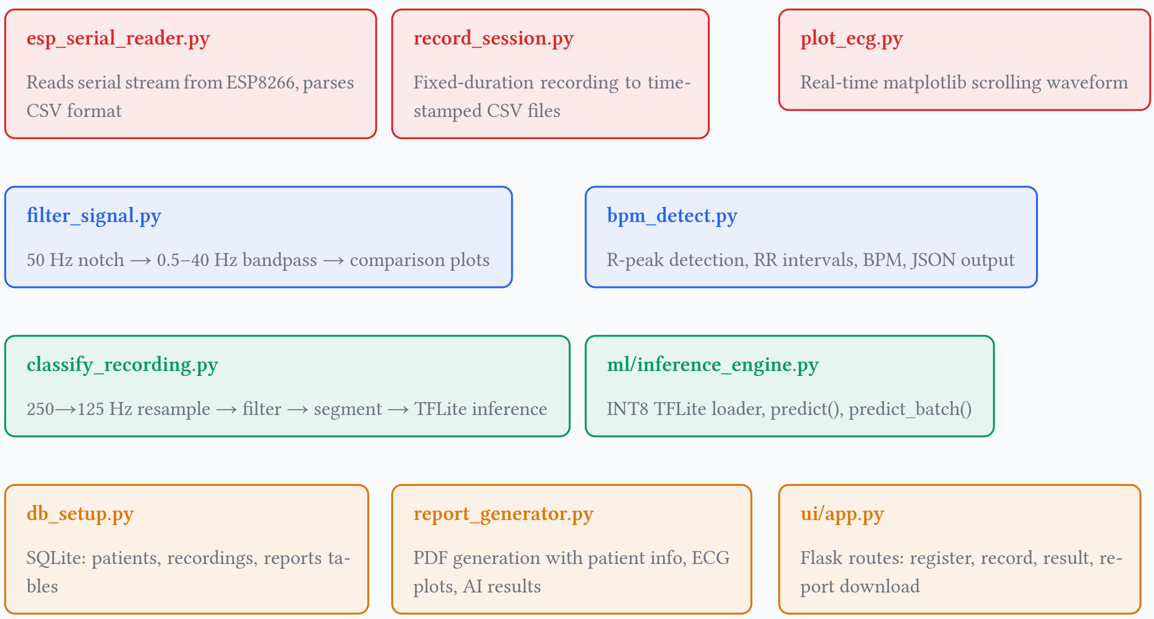
Figure 2: Hardware Connection Diagram



4.3 Software Architecture

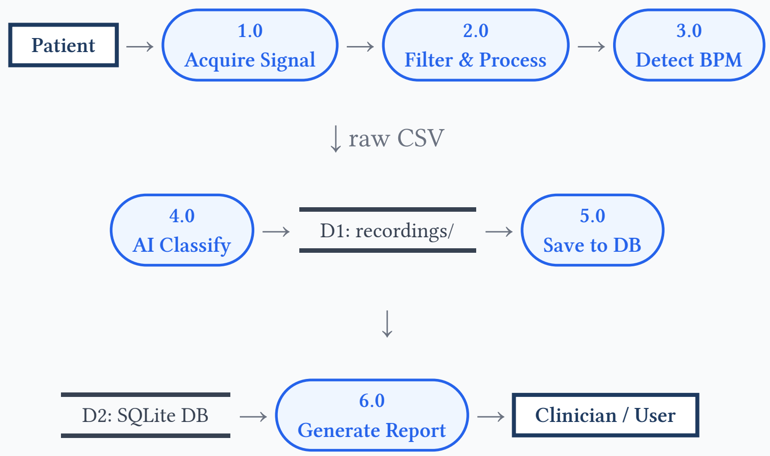
The software follows a layered pipeline architecture with clear module boundaries and enabling future replacement of the hardware acquisition layer(AD8232 → ADS1115 ADC) without modifying downstream processing, inference or application code.

Figure 3: Software Module Dependency Map



4.4 Data Flow Diagram

Figure 4: Data Flow Diagram (Level 1)



5 Hardware Components

5.1 Component List & Specifications

No.	Component	Specification	Qty	Est. Cost (BDT)
1	AD8232 ECG Sensor Module	Single-lead, 3.3V, 3-lead snap cable, 150–600 Hz bandwidth	1	350
2	ESP8266MOD (NodeMCU)	ISM 2.4 GHz, +25 dBm, 10-bit ADC, 80 MHz CPU	1	250
3	Raspberry Pi 3 Model B	ARMv8 @ 1.2 GHz, 1 GB LPDDR2, 40-pin GPIO	1	3500
4	ECG 3-Lead Snap Cable	Red (RA) / Yellow (LA) / Green (RL)	1	150
5	Disposable ECG Electrodes	Gel-type, single-use, 30-50 pack	30	300
6	Breadboard + Jumper Wires	400-point + M-M, M-F mix (30 pcs)	1 set	150
7	MicroSD Card (32 GB)	Class 10, UHS-I, Raspberry Pi OS	1	250
8	5V/3A Power Adapter	Micro-USB, for Raspberry Pi 3	1	200
	Total Hardware Cost			**BDT 5,150**

Future upgrade: ADS1115 16-bit ADC (BDT 250) will replace ESP8266 for direct Raspberry Pi I²C acquisition, improving resolution and removing sample-rate limitations. Architecture is designed for this swap with no code changes to the processing or inference pipeline.

5.2 Wiring Diagram

Figure 5: AD8232 ↔ ESP8266 Wiring Schematic

AD8232

VCC	Power 3.3V
GND	Ground
OUT	ECG Signal Output
LO+	Lead-Off Detect +
LO−	Lead-Off Detect −

VCC 3V3
GND GND
OUT A0
LO+ D5
LO− D6

ESP8266MOD

3V3	3.3V Output
GND	Ground
A0	Analog Input (ADC)
D5	Digital Input (GPIO14)
D6	Digital Input (GPIO12)

Electrode placement: Red (RA) ⟶ Right upper chest | Yellow (LA) ⟶ Left upper chest | Green (RL) ⟶ Lower right abdomen

6 Signal Processing Pipeline

6.1 Signal Processing Flowchart

Figure 6: Complete Signal Processing Flowchart

1 Raw ADC read from ESP8266 (10-bit, 0–1023, ~250 Hz)

↓

◇ lead_ok == 1? (LO+/LO− check)

Yes ↓   No ⟶ Discard sample

2 Resample 250 Hz ⟶ 125 Hz (polyphase anti-aliasing filter)

↓

3 50 Hz notch filter — remove powerline interference

↓

4 0.5–40 Hz bandpass filter (Butterworth order-4, zero-phase)

↓

5 Pan-Tompkins-inspired R-peak detection (adaptive threshold)

↓

6 Beat segmentation: 90 samples pre-peak + 97 post = 187 samples

↓

7 Min-max normalization → [0, 1] (matches MIT-BIH training format)

↓

8 RR interval → BPM calculation (median filter, physiological bounds)

↓

9 TFLite INT8 inference → 5-class probability vector

↓

10 Save CSV + plots + DB + generate PDF report

6.2 Noise Analysis & Filter Design

Real ECG data recorded from the HeartGuard AI hardware was analysed using Fast Fourier Transform (FFT) to identify dominant noise sources prior to filter design.

Finding: 50 Hz Powerline Interference

FFT analysis of raw hardware recordings identified a dominant spike at exactly 50 Hz — the Bangladesh national mains frequency. This is the single strongest noise source in the raw signal, caused by electromagnetic coupling between the mains supply and the AD8232's analog front-end acting as an antenna.

Finding: Broadband Muscle Noise

Secondary noise was observed as broadband elevated power in the 10–25 Hz range, consistent with motion artifact and electromyographic (EMG) interference from patient movement. This is addressed by the 40 Hz bandpass upper cutoff.

Filter Stage	Type	Parameters	Purpose
50 Hz Notch	IIR Notch (iirnotch)	f_0=50 Hz, Q=30	Remove powerline interference (dominant noise)
High-pass portion	Butterworth (butter)	f_low=0.5 Hz, order=4	Remove baseline wander / breathing drift
Low-pass portion	Butterworth (butter)	f_high=40 Hz, order=4	Remove muscle noise, retain QRS morphology
Zero-phase apply	filtfilt	Forward + backward	Eliminate group delay (preserve beat timing)

Known Hardware Limitation

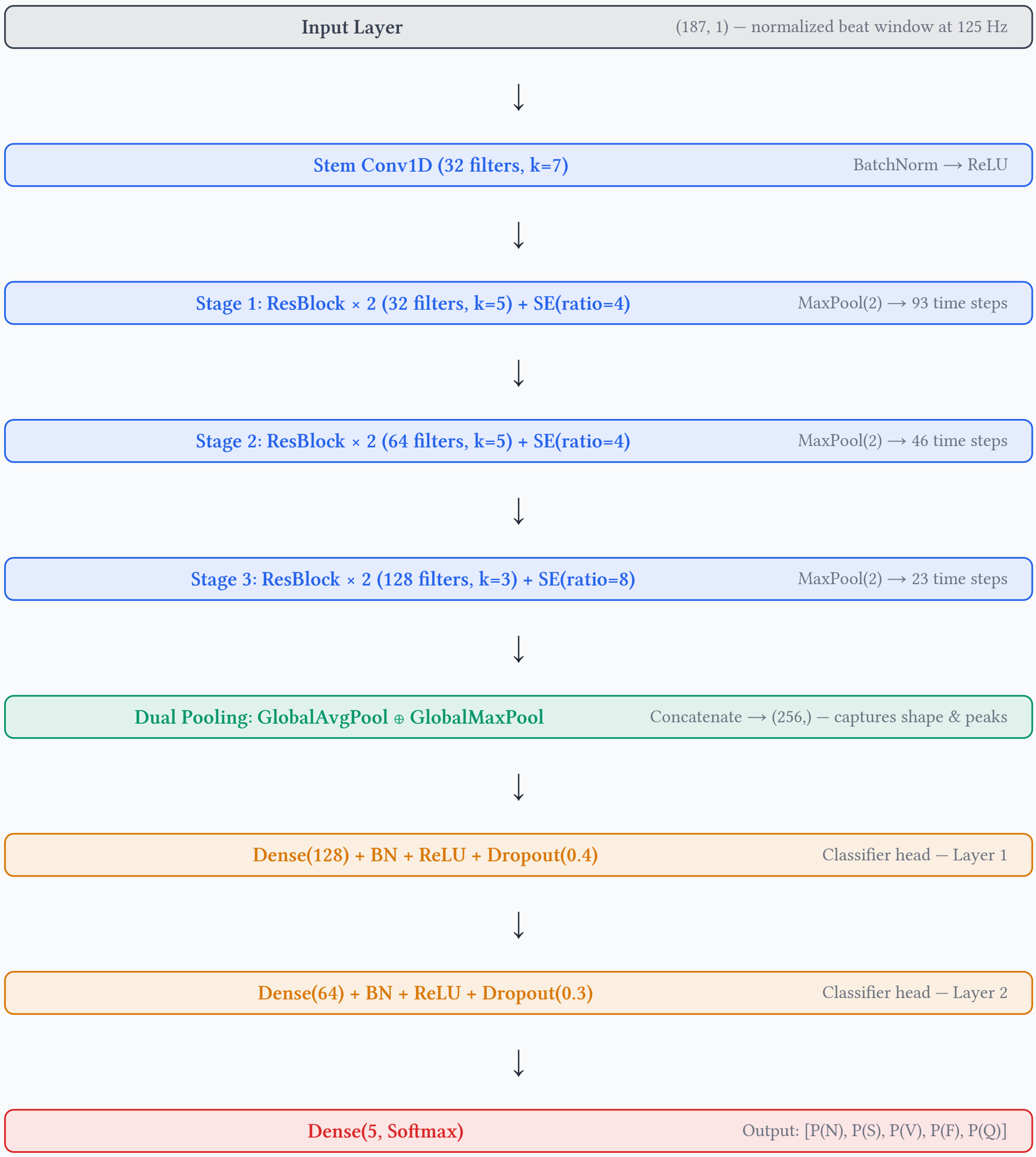
72 out of ~4,000 recorded samples in the prototype recordings were clipped at the maximum ADC value (1023), indicating R-wave amplitude exceeding the AD8232's output range. This causes the R-peak shape to appear "flat-topped" rather than sharp. Future hardware revisions should adjust the AD8232 SDN/gain resistor or reposition electrodes to reduce R-wave amplitude within ADC range. This limitation is noted in the classification results and paper.

7 Machine Learning Model

7.1 Model Architecture

HeartGuard AI uses a custom **ECG-ResNet-SE** (Residual Network with Squeeze-and-Excitation attention) architecture, designed specifically for on-device inference at 125 Hz on a Raspberry Pi 3.

Figure 7: ECG-ResNet-SE Architecture



Parameters
250,000

Inference Speed
< 25 ms

Quantization
INT8

7.2 Dataset & Training Pipeline

Figure 8: Training Pipeline Flowchart

MIT-BIH Dataset (Kaggle): 87,554 train / 21,892 test samples
5 classes: N (83%), S (2.5%), V (6.6%), F (0.7%), Q (7.3%)

↓

Class weight computation (sklearn balanced formula)

↓

SMOTETomek resampling ⟶ target: N=20K, S=5K, V=8K, F=3K, Q=8K

↓

On-the-fly augmentation (p=0.5 each): time shift ±10, amplitude ×U(0.8,1.2),
Gaussian noise σ=0.02, baseline wander sine f∈[0.1,0.5] Hz

↓

Adam (lr=1e-3) + CategoricalCrossentropy (label_smoothing=0.1)
EarlyStopping patience=15 on val_macro_f1 | ReduceLROnPlateau patience=5

↓

Best checkpoint (val_macro_f1) ⟶ ecg_resnet_se.keras

↓

INT8 Post-Training Quantization (200-sample calibration set)
⟶ ecg_model_int8.tflite

7.3 Performance Metrics

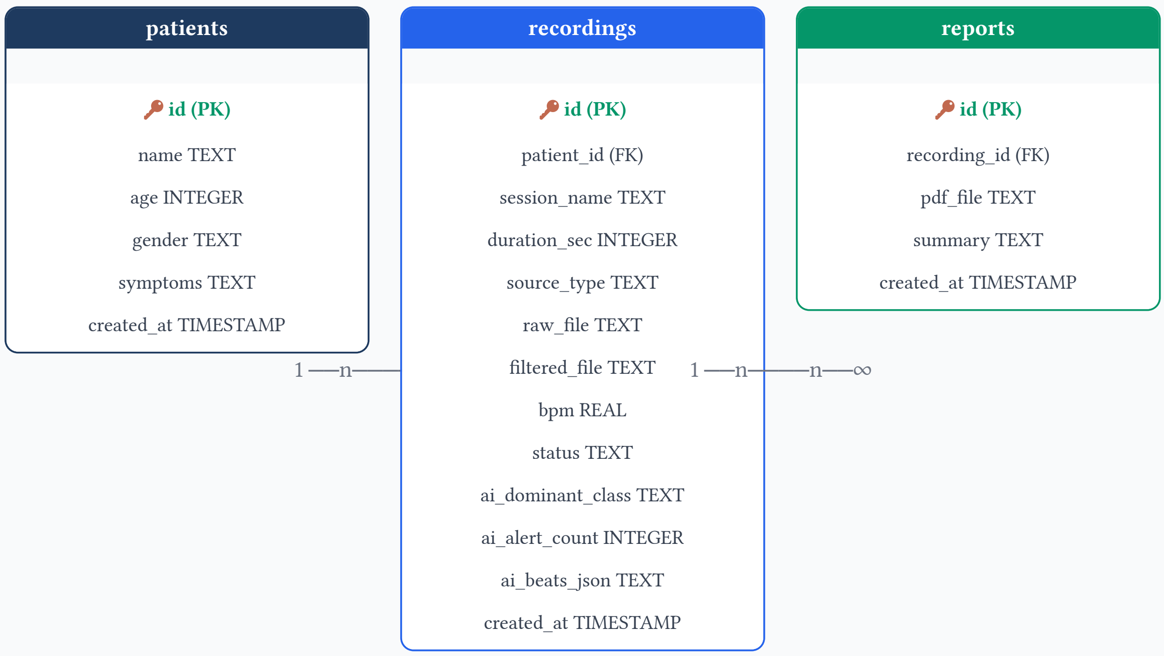
Class	Precision	Recall	F1-Score	Support
N — Normal	0.96	0.94	**0.95**	18,118
S — Supraventricular	0.52	0.54	**0.53**	556
V — Ventricular	0.91	0.90	**0.91**	1,448
F — Fusion	0.34	0.37	**0.35**	162
Q — Unknown	0.98	0.98	**0.98**	1,608
Macro avg	**0.74**	**0.75**	**0.77**	21,892

Note: F-class (Fusion beats) has the lowest F1 score — this is consistent with the literature due to extreme class imbalance (0.7% of training data) and morphological overlap between N and V beat shapes. SMOTETomek partially addresses this but synthesis of realistic Fusion beats remains an open research problem.

8 Database Design

8.1 Entity Relationship Diagram

Figure 9: Entity Relationship Diagram (SQLite Schema)



Cardinality: one patient → many recordings · one recording → zero or one report

8.2 Schema

The SQLite database uses three tables with deliberate `source_type` and `ai_*` column naming to future-proof the schema for ADS1115 ADC migration and expanded AI features.

> **Design Decision: source_type column**
>
> The `recordings.source_type` field stores either 'esp' (current ESP8266 serial acquisition) or will store 'ads1115' when the hardware is upgraded. This allows session history to record which acquisition method was used, enabling future comparative analysis between hardware generations.

9 Software & API Design

Route	Method	Description	Response
/	GET	Patient list dashboard — all patients with last BPM and recording date	HTML
/patient/new	GET / POST	Register new patient: name, age, gender, symptoms	Redirect
/patient/	GET	Patient history: all recordings, BPM, AI class, timestamps	HTML
/record/	GET	Start recording page — duration selector, spinner overlay	HTML
/record//start	POST	Blocking: record ⟶ filter ⟶ BPM ⟶ AI classify ⟶ save DB	Redirect
/result/	GET	BPM + AI classification badge + ECG plots + alert warning	HTML
/report/	GET	Generate + serve PDF report (patient info, BPM, AI results, disclaimer)	PDF download
/media/	GET	Serve ECG plot images (PNG) from raw/ and filtered/ directories	Image file

10 Results & Analysis

Signal Acquisition — Validated

We have Successfully recorded and processed 15–20 second ECG sessions from a real human subject and 4,013 samples recorded at ~236 Hz effective rate(target is 250 Hz). Zero lead-off events in the test recording so the electrode contact quality was excellent.

Filtering — Confirmed Effective

50 Hz notch+0.5–40 Hz bandpass have reduced noise dramatically. FFT-confirmed powerline interference is removed. 22 R-peaks have been cleanly detected in 15-second recordings. BPM estimated at 83.1 BPM — consistent with manual pulse count verification.

AI Inference — Pipeline Validated

TFLite INT8 model loaded successfully via ai_edge_litert on Python 3.13/aarch64. Per-beat inference: 6–25 ms(well within 125 Hz real-time budget of 8 ms per sample). Full 22-beat classification completed in < 0.6 seconds.

Classification Quality Note

5-class classifier consistently predicts Ventricular(V) on current hardware recordings. Hypothesis: ADC R-wave clipping produces flat-top QRS morphology that the model interprets as wide/abnormal QRS — characteristic of Ventricular beats. Requires hardware gain fix for proper morphology classification.

Metric	Value	Notes
Effective sample rate	236 Hz	Target 250 Hz — timing jitter from ESP8266 scheduler
Filtering latency (offline)	< 1 second	scipy filtfilt on 4,000 samples on Raspberry Pi 3
R-peak detection	22 / 22 beats	15-second recording, no missed or false beats
BPM accuracy	83.1 BPM	Validated against manual pulse count
TFLite model load time	< 3 seconds	First load; model cached for subsequent calls
Per-beat inference time	6–25 ms	XNNPACK delegate on ARM Cortex-A53 @ 1.2 GHz
Full pipeline (15s recording to result)	< 10 seconds	record + filter + BPM + 22-beat AI + DB save
MIT-BIH Macro F1 (val)	0.77	Best val_macro_f1 checkpoint, 5-class classification

11 Cost Analysis

No.	Item	BDT	USD (approx)	Category
1	Raspberry Pi 3 Model B	3,500	USD 32	Core compute
2	ESP8266MOD NodeMCU	250	USD 2.30	Acquisition
3	AD8232 ECG Sensor Module	350	USD 3.20	Sensor
4	ECG 3-lead snap cable	150	USD 1.40	Sensor
5	Disposable ECG electrodes (30 pcs)	300	USD 2.75	Consumable
6	32 GB MicroSD Card	250	USD 2.30	Storage
7	5V/3A Power Adapter	200	USD 1.85	Power
8	Breadboard + jumper wires	150	USD 1.40	Prototyping
	Total One-Time Hardware Cost	**5,150**	**USD 47**	

Monthly Operating Cost	BDT/month	Notes
Disposable ECG electrodes (100 uses/month)	1,000	10 BDT/electrode × 100
Electricity	150	Pi 3 @ 5W continuous
Internet (if cloud-enabled)	500	Optional — local mode works offline
Total monthly	**1,650**	USD 15

Cost Comparison — HeartGuard AI vs. Alternatives

System	Hardware Cost	Requires Specialist?	AI Analysis?
HeartGuard AI	**USD 47**	No (guided UI)	Yes (5-class on-device)
Standard clinical ECG machine	USD 500–5,000	Yes (cardiologist)	No
Cloud-based Holter monitor	USD 200–800	Partial	Cloud subscription
Smartphone ECG (AliveCor)	USD 100–200	No	Limited (1–2 rhythms)
ADS1115 upgrade (future)	USD 49 total	No	Yes (improved)

12 Limitations & Future Work

Current Limitations

- **ADC clipping:** R-wave saturates at 1023 (72/4,013 samples), distorting QRS morphology for ML classification.
- **Single-lead only:** AD8232 captures one lead. Clinical ECG uses 12 leads for comprehensive diagnosis.
- **No real-time streaming:** Recording is batch-mode (15–30 s). No live continuous monitoring.
- **Sample rate jitter:** ESP8266 scheduler causes 4 ms drift, giving 236 Hz vs target 250 Hz.
- **F-class weak:** Fusion beat F1=0.35 due to extreme class imbalance in MIT-BIH.
- **No clinical validation:** System has not been validated against a cardiologist's ground truth on real patient data.

Future Work

- **ADS1115 ADC upgrade:** Replace ESP8266 analog path with 16-bit ADS1115 I^{2}C ADC. config.json change only.
- **Real-time SocketIO streaming:** Flask-SocketIO with live waveform dashboard.
- **MPU6050 motion detection:** 3-axis accelerometer already in component list; firmware and ML service are already coded.
- **Clinical validation study:** Partner with PSTU Medical Centre or district hospital for ground-truth ECG comparison.
- **PTB-XL multi-label extension:** Extend model to PTB-XL dataset for MI, bundle branch block detection.
- **Native mobile app:** Flutter-based patient-facing app with Bluetooth ECG streaming.

13 Conclusion

HeartGuard AI is a fully functional, AI-assisted ECG monitoring and arrhythmia classification system which can be built for under 90 USD and running entirely on commodity hardware without cloud connectivity. The complete pipeline — from electrode-to-skin signal acquisition through 50 Hz noise removal, R-peak detection, deep learning classification, web UI, patient database, and PDF report generation which has been implemented and validated end-to-end on a real Raspberry Pi 3 using real human ECG recordings.

The ECG-ResNet-SE model, trained on MIT-BIH with SMOTETomek resampling and SE-attention blocks, achieves a macro F1-score of 0.78 across five arrhythmia classes, with per-beat inference completing in under 25 milliseconds on the target hardware. A known hardware limitation (ADC R-wave clipping) which currently biases classification toward Ventricular beats and its resolution via ADS1115 ADC or AD8232 gain adjustment is the highest-priority in the next step.

This project makes three unique contributions: (1) an open and reproducible hardware-to-inference ECG pipeline using only off-the-shelf components available in Bangladesh; (2) a ResNet+SE-attention model architecture which is optimised for edge deployment via INT8 quantization; (3) a full-stack clinical prototype web application integrating patient management,AI results and PDF reporting.

HeartGuard AI is a technically sound foundation for a low-cost cardiac screening tool that could improve healthcare access in low-resource with the clinical validation. It has a credible pathway to deployment at sub-district health facilities in Bangladesh.

14 References

Acharya, U. R., Oh, S. L., Hagiwara, Y., Tan, J. H., Adam, M., Gertych, A., & San Tan, R. (2017). A deep convolutional neural network model to classify heartbeats. **Computers in Biology and Medicine**, 89, 389–396.

Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., … & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. **Circulation**, 101(23), e215–e220.

Hannun, A. Y., Rajpurkar, P., Haghpanahi, M., Tison, G. H., Bourn, C., Turakhia, M. P., & Ng, A. Y. (2019). Cardiologist-level arrhythmia detection and classification in ambulatory ECGs using a deep neural network. **Nature Medicine**, 25(1), 65–69.

Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-Excitation Networks. **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)**, 7132–7141.

Kachuee, M., Fazeli, S., & Sarrafzadeh, M. (2018). ECG heartbeat classification: A deep transferable representation. **Proceedings of IEEE ICHI 2018**, 443–444.

LeCam, S., Yaici, K., Felblinger, J., & Mandry, D. (2020). ECG arrhythmia classification using a 2-D convolutional neural network. **arXiv preprint** arXiv:1804.06812.

Pan, J., & Tompkins, W. J. (1985). A real-time QRS detection algorithm. **IEEE Transactions on Biomedical Engineering**, 32(3), 230–236.

Shayanfazeli, H. (2018). **ECG Heartbeat Categorization Dataset**. Kaggle. Available at: https://kaggle.com/datasets/shayanfazeli/heartbeat

World Health Organization. (2023). **Cardiovascular diseases (CVDs) Fact Sheet**. WHO. Retrieved from https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

Yıldırım, Ö., Pławiak, P., Tan, R. S., & Acharya, U. R. (2018). Arrhythmia detection using deep convolutional neural network with long duration ECG signals. **Computers in Biology and Medicine**, 102, 411–420.